



2019 BEGINNERS ASSIGNMENT 1

SIYANOQOBA REGIONAL OLYMPIAD TRAINING PROGRAMME

BEGINNERS ASSIGNMENT 1

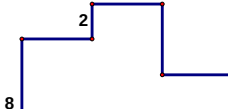
(Final date of submission: 4 March 2019)

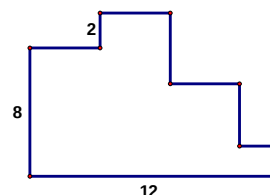
All learners attending Siyanqoba classes are encouraged to submit assignments. All those who submit their solutions will be sent detailed solutions. It will give you invaluable practice for Mathematics Olympiads. This is a great plus. The three assignment marks and your performance in Rounds 1 and 2 will determine whether you will be selected to participate in an all expenses paid Mathematics Camp in Stellenbosch in December.

This is the first assignment of three. Each assignment has ten questions. PLEASE WRITE YOUR PERSONAL DETAILS AND ANSWERS CLEARLY IN THE ANSWER SHEET on the last page of the assignment. Scan and email the ONLY the answer sheet to

siyangoba.beginner.assignments@gmail.com

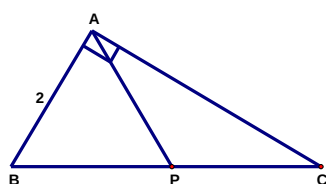
Alternatively, you may hand your answer sheet to your coach who will scan and send it to me. As a last resort, fax it to 031 260 7806.

- [3 marks]** If $a * b = ab + \frac{a}{b}$ calculate $7 * 3$.
- [4 marks]** A hiker walks 1 km East, then 2 km North, then 3 km West, then 4 km South, then 5 km East and finally 6 km North. What is the hiker's straight-line distance in km from the starting point?
- [4 marks]** A cyclist races around a track for three hours, taking two minutes and thirty seconds for each lap. How many laps did she complete?
- [4 marks]** The polygon in the figure has sides which meet at right angles. Side lengths are as shown. Find the perimeter of the polygon.



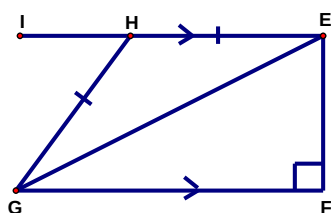
5. **[6 marks]** The sequence 123456789123456789....is continued until the sum of all the digits is 388. What is the last digit in the sequence?

6. [5 marks]



$\triangle ABC$ is right-angled at A, and $\triangle ABP$ is equilateral with $AB = 2$. Determine the length of AC.

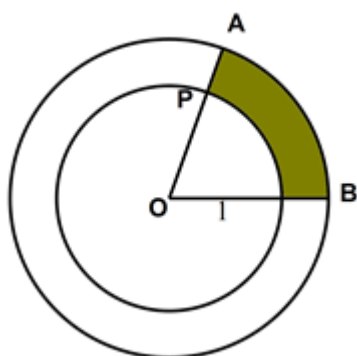
7. [6 marks]



Let $x = \angle IHG$.
Calculate $\angle FEG$ in terms of x .

8. [5 marks] The decimal form of the number $\frac{3}{13}$ is the recurring decimal 0,230769230769... What is the digit in the 2019th place after the decimal point?

9. [6 marks]



Two circles have the same centre O, and the radius of the smaller one is 1. The radii OA and OB are drawn so that
area of the shaded region : area of the sector OAB = 7 : 16.
What is the length of PA?

10. [7 marks] If $a + b = 6$ & $ab = 4$ determine $a^4 + b^4$.

ANSWER SHEET: ASSIGNMENT 1 (DUE DATE: 4 MARCH, 2019)

Siyanqoba Centre (Circle one of the following:

UP TUT UKZN BLM EMS RHODES UL UNIVEN/TENGWE WITS
ALGOA SPU NW

Surname:

Name:

Gender:

Race:

Grade:

School:

Cell no:

E-mail address:

QUESTION	YOUR ANSWER	MAXIMUM MARK
1		3
2		4
3		4
4		4
5		6
6		5
7		6
8		5
9		6
10		7
TOTAL		50

Send to: siyanqoba.beginner.assignments@gmail.com